

## Lecture 14

### Recall

Thm [Comb. Null.] •  $f$  non-zero  $n$ -var. poly on  $\mathbb{F}_p^n$

$$\bullet S_1, \dots, S_n \subseteq \mathbb{F}_p$$

if  $f$  has a term  $x_1^{k_1} \dots x_n^{k_n}$  w/  $\begin{cases} \sum k_i = \deg f \\ k_i < |S_i| \end{cases} \forall i \in [n]$

$\Rightarrow f$  cannot vanish on  $S_1 \times \dots \times S_n$ .

• How do we use it to lower bound  $|A|$ ?

Strategy • Supp.  $|A|$  too small

$\Rightarrow$  find a polyn  $f$  w/ "low degree"  
(w.r.t. some  $S_i$ )  
s.t.  $f$  vanishes on too large prod. set  $(\prod S_i)$ .

Q: How many affine hyperplanes we need to cover the hypercube (apart from origin:  $\{0,1\}^n \setminus \{0\}$ )

Ans if not affine, two suffices.

Thm [Alon-Furedi 93] Let  $H_1, \dots, H_m$  be hyperplanes in  $\mathbb{R}^n$  s.t. none of them contain  $0$  and

they together cover all other vertices of the hypercube  $\{0,1\}^n \setminus \{0\}$ .

Thought process

• Natural prod. set  $\{0,1\}^n$

$$\Rightarrow S_i = \{0,1\}$$

- $|S_i| = 2 \Rightarrow$  suggest  $\bullet x_1^1 x_2^1 \dots x_n^1$  high order term
- $\deg f = n$

Need  $f$  to vanish on  $\{0,1\}^n = \prod S_i$

(\*)  $\forall j \in [m]$  Write

$$H_j := \{x : \langle a_j, x \rangle = b_j\}, \text{ for some } a_j, b_j \neq 0$$

- Known:  $\forall x \in \{0,1\}^n \setminus \{0\}, \exists j$  s.t.  $x \in H_j$  i.e.  $b_j - \langle a_j, x \rangle = 0$

Suggest  $\Rightarrow \prod_{j=1}^m (b_j - \langle a_j, x \rangle)$  vanish on hypercube apart from 0.

- need to add to above a poly  $\begin{cases} -\prod_{j=1}^m b_j & \text{at } 0 \\ 0 & \text{elsewhere in hypercube} \end{cases}$

$$\left(-\prod_{j=1}^m b_j\right) \cdot \prod_{i=1}^n (1-x_i)$$

Pf: • Write  $H_j$  as in (\*)

$$\text{Let } f(x) = \underbrace{\prod_{j=1}^m (b_j - \langle a_j, x \rangle)}_{\text{vanishes on } H_j} - \underbrace{\left(\prod_{j=1}^m b_j\right) \cdot \prod_{i=1}^n (1-x_i)}_{\text{vanishes at } 0}$$

$$S_1 = \dots = S_n = \{0,1\}$$

Supp.  $m < n \Rightarrow \deg f = \max\{m, n\} = n$

CHECK coeff. of  $x_1 \dots x_n$  is non-zero ✓😊

but  $f$  vanishes on  $\prod S_i \Rightarrow$  comb. null. 

Thm [Chevalley - Warning 1935]

- $p$  prime
  - $f_1, \dots, f_m$  polyn on  $\mathbb{F}_p^n$   
w/  $\sum_{i=1}^m \deg f_i < n$
- $\Rightarrow$  if  $\exists$  a common root.  
 $\Rightarrow \exists$  other common root.

Remark: Original statement stronger: # common roots is divisible by  $p$ . In particular, if  $\exists$  one  $\Rightarrow \geq p$  common roots.

Thought process: Natural prod:  $\mathbb{F}_p^n \Rightarrow \forall i \in [n] S_i = \mathbb{F}_p$   
• supp  $f_j$ 's have exactly one common root  $c \in \mathbb{F}_p^n$ .

$\forall c' \neq c, \exists j, f_j(c') \neq 0$  in  $\mathbb{F}_p$

$\Rightarrow$  take  $\prod_{j=1}^m (1 - f_j(x)^{p-1}) = \begin{cases} 0 & \mathbb{F}_p^n \setminus c \\ 1 & c \end{cases}$

To make it vanish also at  $c$ , add to it

$$- \prod_{i=1}^n (1 - (x_i - c_i)^{p-1})$$

Pf: Take  $S_1 = \dots = S_n = \mathbb{F}_p$  &

$$f(x) = \prod_{j=1}^m (1 - f_j(x)^{p-1}) - \prod_{i=1}^n (1 - (x_i - c_i)^{p-1})$$

$$\begin{aligned} \deg f &= \max \left\{ (p-1) \sum_{j=1}^n \deg f_j, (p-1)n \right\} \\ &= (p-1)n \end{aligned}$$

•  $x_1^{p-1} \dots x_n^{p-1}$  coeff. non-zero

but  $f$  vanishes on  $\mathbb{F}_p^n = \prod S_i$  C.N.  $\hookrightarrow \square$

### § Finding regular subgraph

of  $n$ -vtx graph  $G$

• Thm [Pyber 85]  $\forall k \in \mathbb{N}$ ,  $d(G) \geq 32k^2 \log n$

$\Rightarrow \exists$   $k$ -reg. subgraph.

• Thm [Pyber - Rödl - Szemerédi 95]

•  $d(G) \geq c_k \log \Delta(G) \Rightarrow \exists$   $k$ -reg subgr.

•  $\exists G$  w./  $d(G) \geq c \cdot \log \log n$  w./ no  
3-reg. subgr.

Thm [Alon - Friedland - Kalai 1984]

Let  $p$  be a prime,  $G$  be a loopless multigraph

w./  $\begin{cases} d(G) > 2p-2 \\ \Delta(G) \leq 2p-1 \end{cases} \Rightarrow \exists$   $p$ -reg. subgr.

- A standard trick to view a graph algebraically is to identify  $G$  w./ its edge set  $E(G)$

View  $E(G) \xleftrightarrow{\text{a vect in}} \{0,1\}^{\binom{n}{2}}$

- subgr. of  $G \xrightarrow{\text{vect}} \{0,1\}^{E(G)}$



- $\exists$   $p$ -reg subgr  $\Leftrightarrow$  a vect  $x \in \{0,1\}^m \setminus \{0\}$

$\forall v \in V(G) \quad m = E(G)$   
 s.t.  $\sum_{e \ni v} x_e \equiv 0 \pmod p$

Supp. no such vector exists

Thought process

- prod. set  $\prod_{e \in E(G)} S_e, S_e = \{0,1\}$

- high order term  $\prod_{e \in E(G)} x_e \Rightarrow \deg f = m$

Want  $f$  vanishes on  $\{0,1\}^m$

- $\forall x \in \{0,1\}^m \setminus \{0\}, \exists v \in V(G) \text{ s.t. } \sum_{e \ni v} x_e \not\equiv 0 \pmod p.$

$\Rightarrow$  suggest  $\prod_{v \in V(G)} (1 - (\sum_{e \ni v} x_e)^{p-1}) = \begin{cases} 1, & x=0 \\ 0, & \{0,1\}^m \setminus \{0\} \end{cases}$

add to it  $-\prod_{e \in E(G)} (1 - x_e) = \begin{cases} -1, & x=0 \\ 0, & \{0,1\}^m \setminus \{0\} \end{cases}$

Pf: Let  $f(x) = \prod_{v \in V(G)} (1 - (\sum_{e \ni v} x_e)^{p-1}) - \prod_{e \in E(G)} (1 - x_e)$

$\deg f = \max \{ n(p-1), m \} = m$

$$\text{as } d(G) = \frac{2m}{n} \geq 2p-2.$$

- highest order term  $\prod_{e \in E(G)} x_e$  w/ non-zero coeff.  
 $\hookrightarrow$  C.N.  $\textcircled{M}$

§ Finding zero-sum multisubset.

[Permanent lem] Let  $A$  be an  $n \times n$  matrix

over a field  $\mathbb{F}$  w/  $\text{per}(A) \neq 0$ .

Let  $b = (b_1, \dots, b_n) \in \mathbb{F}^n$ ,  $S_1, \dots, S_n \subseteq \mathbb{F}$  w/  $|S_i| = 2$ .

$\Rightarrow \exists x \in \prod_{i=1}^n S_i$  s.t.  $Ax - b$  has no zero coord.

i.e.  $\forall i \in [n] \quad (Ax)_i \neq b_i$

Exer!

Thm (Erdős - Ginzburg - Ziv)

$p$  prime,  $\forall$  A multiset of  $\mathbb{Z}_p$  size  $2p-1$

$\Rightarrow \exists$  sub-multiset of size  $p$  whose elements sum up to  $0 \pmod p$ .